SPLINES AND CURVES 6

CS 184: FOUNDATIONS OF COMPUTER GRAPHICS

1 A Polynomial Interpolation

Our goal is to fit a polynomial to given points and derivatives of a curve. We can solve this problem by formulating it as a system of linear equations in the coefficients of the polynomial.

1. List all degree 2 polynomials satisfying: $f(0) = 1$, $f(1) = 2$, $f(2) = 5$.

2. How many degree 3 polynomials satisfy the given constraints?

3. Suppose we have a list of constraints:

$$f(0) = p_0,\ f'(0) = d_0,\ f(1) = p_1,\ f'(1) = d_1,\ \ldots,\ f(k) = p_k,\ f'(k) = d_k.$$

   Since we have $2(k + 1)$ constraints (one function and one derivative condition per point), the unique interpolating polynomial must have degree $2k + 1$.

   For a function f, what are the tradeoffs when either

   - solving for a single degree $2k + 1$ polynomial, versus
   - taking the point and derivative constraints at i and $i - 1$ for $i = 1, \ldots, k$ and using them to fit k cubic Hermite splines?

4. Consider a cubic polynomial $f(t) = at^3 + bt^2 + ct + d$ that satisfies the following conditions:

$$\begin{aligned} f(0) &= f_0, \\ f(1) &= f_1, \\ f''(0) &= f_0'', \\ f''(1) &= f_1''. \end{aligned}$$

Write the matrix that, when inverted and applied to the vector $(f_0, f_1, f_0'', f_1'')^T$, allows you to recover the coefficients a, b, c, and d of the polynomial.

5. Given the numerical inverse of the matrix:

$$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 0 & 0 & -1/6 & 1/6 \\ 0 & 0 & 1/2 & 0 \\ -1 & 1 & -1/3 & -1/6 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} f_0 \\ f_1 \\ f_0'' \\ f_1'' \end{pmatrix}$$

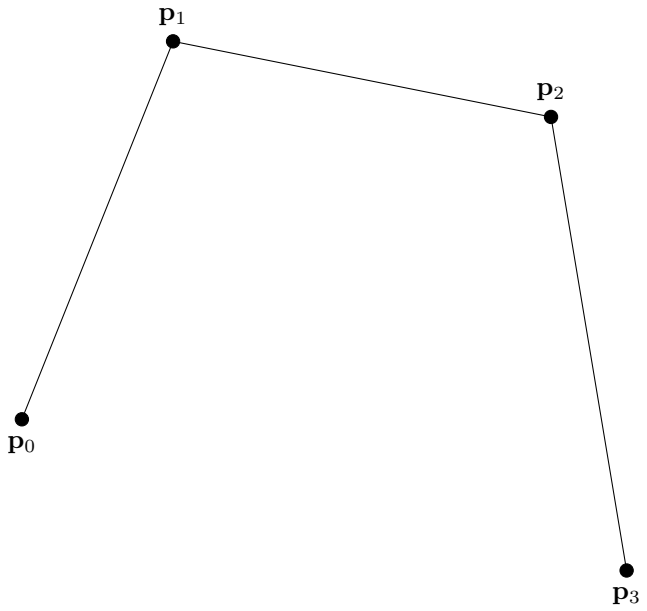
what are the **basis polynomials** that allow expressing $f(t)$ in terms of f_0, f_1, f_0'',, f_1''?

2 de Casteljau's algorithm

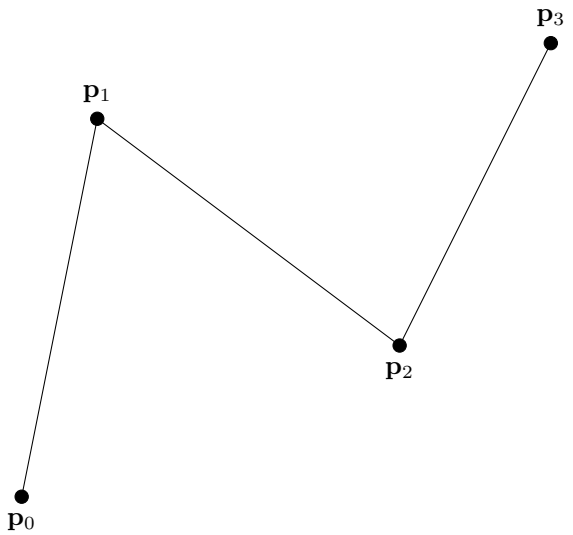
Given $k+1$ points $\mathbf{p_0}, \ldots, \mathbf{p_k}$, create a new set of k points $\mathbf{p'_0}, \ldots, \mathbf{p'_{k-1}}$ by computing $\mathbf{p'_i} = \text{lerp}(\mathbf{p_i}, \mathbf{p_{i+1}}, t)$, where $\text{lerp}(\mathbf{p_i}, \mathbf{p_{i+1}}, t) = (1-t)\mathbf{p_i} + t\mathbf{p_{i+1}}$.

Perform k times to yield a single point, $f(t)$.

1. Use de Casteljau's algorithm to construct $f(1/2)$ with the given control points.



2. Use de Casteljau's algorithm to construct $f(1/3)$ with the given control points.



3. Show that the point with parameter t on the Bézier curve with control points $\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$ is given by $s^3\mathbf{p}_0 + 3s^2t\mathbf{p}_1 + 3st^2\mathbf{p}_2 + t^3\mathbf{p}_3$, where $s = 1 - t$. (Hint: apply de Casteljau's algorithm algebraically to the control points. With this setup, linear interpolation between two points $\mathbf{q_0}$ and $\mathbf{q_1}$ looks like $s\mathbf{q_0} + t\mathbf{q_1}$.)

4. What is this matrix product? (Hint: *don't* expand it. Instead, think about what each matrix in the product does. How are they related to de Casteljau's algorithm?)

$$\begin{pmatrix} s & t \end{pmatrix} \begin{pmatrix} s & t & 0 \\ 0 & s & t \end{pmatrix} \begin{pmatrix} s & t & 0 & 0 \\ 0 & s & t & 0 \\ 0 & 0 & s & t \end{pmatrix}$$

5. For a Bézier curve defined by 3 control points, what is the degree of the polynomial that results from de Casteljau's algorithm? What about for k points?

[1]Here, we consider the control points $\mathbf{p}_j$ to be variables and the resulting function to be a polynomial in t. If you can fix $\mathbf{p}_j$'s, it is possible to obtain a polynomial with lower degree. For example, if every $\mathbf{p}_j = \mathbf{0}$, then the resulting polynomial is $\mathbf{0}$.